

测试时计算综述：从直觉推理到审慎推理

YIXIN JI, Soochow University, China
JUNTAO LI, Soochow University, China
YANG XIANG, Soochow University, China
HAI YE, National University of Singapore, Singapore
KAIXIN WU, Ant Group, China
KAI YAO, Ant Group, China
JIA XU, Ant Group, China
LINJIAN MO, Ant Group, China
MIN ZHANG, Soochow University, China

o1 模型在复杂推理中的卓越性能表明，测试时计算扩展能够进一步释放模型潜力，实现强大的系统 2 思维。然而，目前仍缺乏对测试时计算扩展的全面综述。我们将测试时计算的概念追溯至系统 1 模型。在系统 1 模型中，测试时计算通过参数更新、输入修改、表示编辑和输出校准来应对分布偏移，并提升鲁棒性与泛化能力。在系统 2 模型中，它通过重复采样、自我纠正和树搜索来增强模型解决复杂问题的推理能力。本综述按照系统 1 到系统 2 思维的演变趋势进行组织，突出测试时计算在从系统 1 模型向弱系统 2 模型、再到强系统 2 模型过渡中的关键作用。我们还指出了前沿课题与未来方向。¹

CCS 概念： • 计算机方法论→Artificial 智能；自然语言处理；自然语言生成。

补充关键词与短语：大语言模型，测试时计算，推理

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¹https://github.com/Dereck0602/Awesome_Test_Time_LLMs.

作者联系信息：Yixin Ji, jiyixin169@gmail.com, 苏州大学, 中国苏州; Juntao Li, 苏州大学, 中国苏州, ljt@suda.edu.cn; Yang Xiang, 苏州大学, 中国苏州; Hai Ye, 新加坡国立大学, 新加坡; Kaixin Wu, 蚂蚁集团, 中国杭州; Kai Yao, 蚂蚁集团, 中国杭州; Jia Xu, 蚂蚁集团, 中国杭州; Linjian Mo, 蚂蚁集团, 中国杭州; Min Zhang, 苏州大学, 中国苏州, minzhang@suda.edu.cn.

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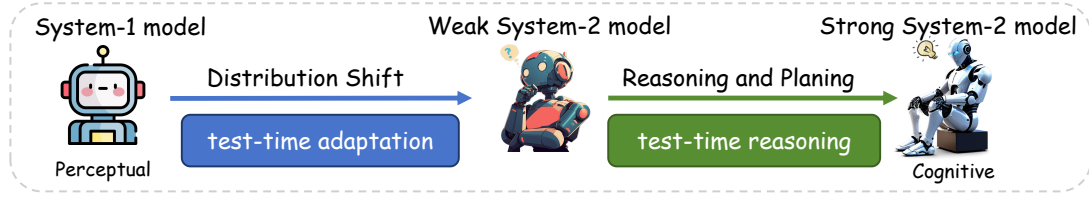


图 1. 系统 1 与系统 2 模型中测试时计算的示意图。

1 Introduction

Over the past decades, deep learning with its scaling effects has been the driving engine behind the artificial intelligence revolution. Particularly in the text modality, large language models (LLMs) represented by the GPT series [24, 226, 229] have demonstrated that larger models and more training data lead to better performance on downstream tasks. However, on the one hand, further scaling in the training phase becomes difficult due to the scarcity of data and computational resources [299]; on the other hand, existing models still perform far below expectations in terms of robustness and handling complex tasks. These shortcomings are attributed to the model’s reliance on fast, intuitive System-1 thinking, rather than slow, deep System-2 thinking [328]. Recently, large reasoning models (LRMs), represented by OpenAI-o1/o3 [227]、DeepSeek-R1 [95] 和 Gemini 2.5 [57] 等配备系统 2 思维模型，因其 outstanding performance in complex reasoning tasks. It demonstrates a test-time compute scaling effect: the greater the computational effort in the inference, the better the model’s performance.

The concept of test-time compute emerged before the rise of LLMs and was initially applied to System-1 models (illustrated in Figure 1). These System-1 models can only perform limited perceptual tasks, relying on patterns learned during training for predictions. As a result, they are constrained by the assumption that training and testing are identically distributed and lack robustness and generalization to distribution shifts [423]. Many works have explored 测试时自适应 (TTA) 通过更新参数 [302, 363]、修改输入 [61]、编辑表示 [247] 和校准输出 [402] 来提升模型鲁棒性。借助 TTA，系统 1 模型放慢其思考过程，并获得更好的泛化能力。然而，TTA 是一种隐式的慢思考，无法像人类那样展现显式的、逻辑性的思考过程，且难以处理复杂的推理任务。因此，启用 TTA 的模型表现出较弱的系统 2 思维。

目前，配备思维链 (CoT) 提示 [326] 的先进大语言模型已使语言模型能够进行显式的系统 2 思维 [97]。然而，普通 CoT 受限于错误累积和线性思维模式 [271]，难以完全模拟人类的非线性认知过程，如头脑风暴、反思和回溯。为了实现更强的系统 2 模型，研究者采用测试时计算策略来扩展模型推理的广度、深度和准确率，例如重复采样 [50]、自我纠正 [263] 和树搜索 [359]。

重复采样模拟了人类思维的多样性，自我纠正使大语言模型能够进行反思，而树搜索则增强了推理的深度和回溯能力。

据我们所知，本文是首篇系统综述测试时计算方法，并深入探讨其在推动模型从系统 1 思维向弱系统 2 思维，最终向强系统 2 思维演进中关键作用的论文。在第 2 节中，我们介绍系统 1 与系统 2 思维的背景。第 3 节和第 4 节分别详述系统 1 和系统 2 模型的测试时计算方法。随后，我们在第 5 节和第 6 节讨论前沿主题与未来方向。此外，我们在第 7 节回顾相关基准。

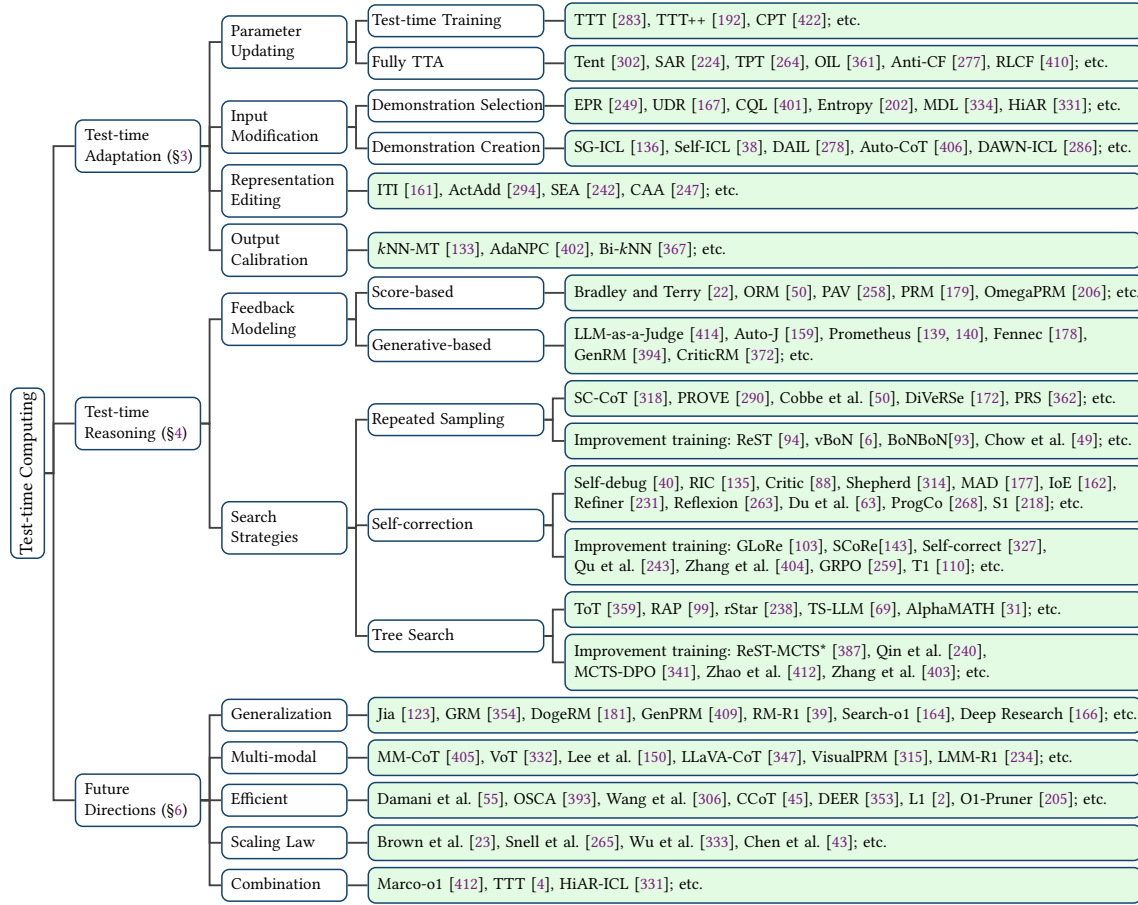


图 2. 测试时计算方法与未来方向的分类体系。

2 背景

系统 1 与系统 2 思维是心理学概念 [127]。当识别熟悉模式或处理简单问题时，人类往往凭直觉作出反应。这种自动、快速的思维被称为系统 1 思维。相比之下，在处理数学证明或逻辑推理等复杂问题时，需要深入而审慎的思考，这被称为系统 2 思维——缓慢且反思性的。在人工智能领域，研究者也用这些术语来描述不同类型的模型 [148]。系统 1 模型基于内部编码的感知信息和世界知识直接作出响应，不展示任何中间决策过程。相比之下，系统 2 模型显式地生成推理过程并逐步解决问题。在大语言模型兴起之前，系统 1 模型是人工智能的主流。尽管许多深度学习模型，如残差网络、变换器和双向编码器表示变换器，在计算机视觉和自然语言处理的各类任务中取得了优异的性能，但这些系统 1 模型类似于人类直觉，缺乏足够的鲁棒性且容易出错 [62, 82, 317]。如今，大语言模型强大的生成与推理能力使得构建系统 2 模型成为可能。魏等人 [326] 提出了思维链，使大语言模型在推理过程中逐步生成中间推理步骤。实证与理论

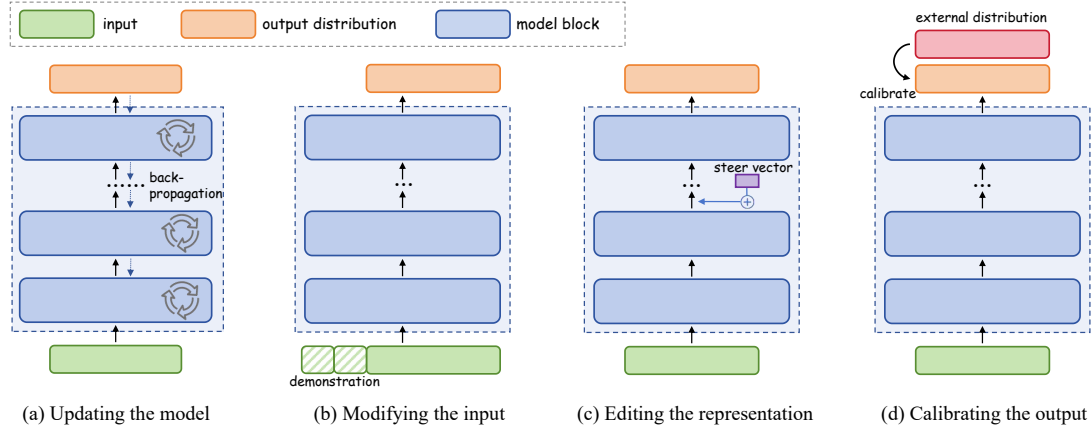


图 3. 各类测试时自适应方法示意图。

results show that this approach significantly outperforms methods that generate answers directly [67, 142, 174, 416]. However, current System-2 models represented by CoT prompting still have shortcomings. The intermediate processes generated by LLMs may contain errors, leading to cumulative mistakes and ultimately resulting in incorrect answers. Although retrieval-augmented generation (RAG) helps mitigate factual errors [91, 292, 324], their impact on improving reasoning abilities remains limited. As a result, CoT-enabled LLMs are still at the weak system-2 thinking stage.

3 面向系统 1 思维的测试时自适应

3.1 更新模型

模型更新利用测试样本信息在推理阶段微调模型参数，使模型适应测试分布。关键在于获取测试分布信息以提供学习信号，并使用合适的参数和优化算法实现高效稳定的更新。

学习信号。在推理阶段，测试样本的真实标签不可用。因此，许多工作尝试设计无监督或自监督目标作为学习信号。根据训练过程是否可修改，现有学习信号可分为两类：测试时训练和完全测试时自适应。测试时训练假设用户可以修改训练过程，引入分布偏移感知的辅助任务。

在测试时自适应过程中，辅助任务损失作为优化的学习信号。许多自监督任务已被证明在图像模态中作为辅助任务是有效的，例如旋转预测 [283]、元学习 [18]、掩码自编码 [74] 和对比学习 [30, 192]。其中，对比学习因其在模态内和跨模态的自监督学习泛化能力，已成功应用于视觉语言任务的测试时自适应 [422]。

相比之下，FTTA 无需访问训练过程，而是利用测试样本的内部或外部反馈作为学习信号。不确定性是最常用的学习信号，其动机是当测试样本偏离训练分布时，模型对其预测的置信度较低，从而导致更高的不确定性。Tent[302] 使用模型预测的熵作为不确定性的测度，并通过最小化熵来更新模型。MEMO[395] 对单个测试样本进行数据增强，然后最小化其

边际熵，在单样本 TTA 设置下比 Tent 更稳定。然而，最小化熵也存在缺陷，盲目降低预测不确定性可能导致模型崩溃并做出平凡预测 [235, 277, 408]。一些工作为最小化熵提出了新的正则化项以避免模型崩溃，包括 Kullback-Leibler 发散 [277]、矩匹配 [102] 和熵匹配 [17]。对于特定任务，少量人类反馈或外部模型奖励也可以作为高质量的学习信号。Gao 等人 [76] 和 Li 等人 [175] 利用用户反馈来适应问答模型。Zhan 等人 [384] 通过使用 COMET[246] 评估翻译质量，将测试时自适应应用于多语言机器翻译任务。在图文检索和图像描述生成等跨模态任务中，RLCF[410] 通过使用 CLIP 分数 [244] 作为 TTA 信号证明了其有效性。在语言建模中，在测试时使用相关上下文文本进行训练可以降低困惑度 [101, 321]。

Hübortter 等人 [121] 从理论上证明它降低了测试样本的不确定性，并提出了一种更有效的主动学习选择策略。

更新参数。为了推动测试时自适应（TTA）在实际场景中的应用，研究人员必须应对效率和稳定性的挑战。为了提高效率，许多方法仅微调一小部分参数，例如归一化层 [255]、软提示 [66, 102, 151, 212, 223, 264]、低秩模块 [112, 113, 122]、适配器模块 [111, 219, 277] 和跨模态投影器 [410]。尽管需要微调的参数数量减少了，TTA 仍然需要额外的反向传播。通常，反向传播的时间成本大约是前向传播的两倍。因此，Niu 等人 [223] 提出了 FOA，它通过协方差矩阵自适应进化策略来调整软提示，从而摆脱了反向传播。

TTA 的稳定性主要体现在两个方面。一方面，无监督或自监督学习信号不可避免地会在优化过程中引入噪声，导致 TTA 沿着错误的梯度方向优化模型。为了解决这个问题，Niu 等人 [224] 和 Gong 等人 [86] 提出了噪声数据过滤策略和鲁棒的锐度感知优化器。另一方面，在实际场景中，测试样本的分布可能会持续变化，但持续的 TTA 优化可能导致模型对原始知识的灾难性遗忘。

情景式 TTA[264, 410] 是一种避免遗忘的设置，它在对单个测试样本进行 TTA 后将模型参数重置为原始状态。然而，情景式 TTA 频繁加载原始模型，导致更高的推理延迟，同时也限制了模型的增量学习能力。为了克服这一困境，一种常见的技巧是指数移动平均 [329, 361]，它融合了先前模型状态的信息。

3.2 修改输入

对于大语言模型（LLM）而言，庞大的参数数量使得基于模型更新的 TTA 方法面临更严峻的效率与稳定性困境。因此，不依赖参数更新的基于输入修改的方法已成为 LLM 中 TTA 的主流方法。输入修改型 TTA 的有效性源于 LLM 的上下文学习（ICL）能力，通过在测试样本前添加一些示例可以显著提升性能。ICL 对示例的选择和顺序高度敏感。因此，输入修改型 TTA 的核心目标是为测试样本选择合适的示例，并以最优顺序排列，以最大化 ICL 的效果。

首先，实证研究 [186] 表明，演示与测试样本越相似，上下文学习（ICL）的性能就越好。因此，使用 BM25 和 SentenceBERT 等检索模型来检索与测试样本语义最接近的演示，并按相似度降序排列 [207, 239]。为了提高演示检索的准确率，Rubin 等人 [249] 和 Li 等人 [167] 通过对比学习专门训练演示检索器。

6 Ji 等人。

学习。随后，随着研究者对上下文学习机制的深入探索，上下文学习被认为是在演示上执行隐式梯度下降 [53]。因此，从训练数据的角度来看，演示也需要具有信息量和多样性 [168, 276]。Wang 等人 [319] 将语言模型视为主题模型，并将演示选择问题形式化为求解贝叶斯最优分类器。此外，示例的排序是另一个重要的改进方向。Lu 等人 [202] 和 Wu 等人 [334] 分别以信息论为指导，选择具有最大局部熵和最小描述长度的示例进行排序。Scarlatos 和 Lan[254] 以及 Zhang 等人 [401] 考虑了演示之间的顺序依赖关系，将其建模为顺序决策问题，并通过强化学习优化演示的选择和排序。

另一类工作 [38, 136, 209, 406] 认为，在实践中，结合有限的外部提供示例可能并不总是最优选择。大语言模型（LLM）可以利用其生成和标注能力来创建更好的演示。DAIL[278] 构建了一个演示记忆库，将之前的测试样本及其预测存储为后续样本的候选演示。DAWN-ICL[286] 进一步将测试样本的遍历顺序建模为规划任务，并通过蒙特卡洛树搜索（MCTS）进行优化。

3.3 编辑表示

对于生成式大语言模型，一些研究发现性能瓶颈不在于编码世界知识，而在于中间层信息与输出之间的巨大差距。在推理阶段，编辑表示有助于将中间知识外化到输出中。PPLM[56] 在小型语言模型的指导下执行基于梯度的表示编辑，以控制输出的风格。

激活加法（ActAdd）[294] 选取两个语义对比的提示，并计算其表示之间的差异作为引导向量，随后将该向量加入残差流。基于对比提示的表示编辑已在更广泛的场景中证明了其有效性，包括指令遵循 [274]、缓解幻觉 [9, 161]、降低毒性 [188, 200] 以及个性 [26]。语义编辑方法（SEA）[242] 将表示投影到与正提示协方差最大、与负提示协方差最小的方向上。他们还引入了非线性特征变换，使表示编辑能够超越线性可分的表示。斯卡莱纳等人

[253] 对引导强度的选择进行了深入研究。他们发现，对每个输出标记施加逐渐减小的引导强度，可以在不降低质量的前提下改善对生成的控制。

3.4 校准输出

利用外部信息校准模型的输出分布也是一种高效且有效的测试时自适应方法 [134]。自适应最近质心分类器（AdaNPC）[402] 设计了一个记忆池来存储训练数据。在推理过程中，给定一个测试样本，AdaNPC 从记忆池中召回 k 个样本，并使用 k 近邻分类器预测测试样本。随后，它将测试样本及其预测标签存入记忆池。随着时间的推移，记忆池中的样本分布逐渐与测试分布对齐。在自然语言处理中，此类方法最具代表性的应用是 k 近邻机器翻译（ k NN-MT）。 k NN-MT [133] 构建了一个数据存储，用于存储上下文表示及其对应的目标标记。在翻译推理过程中，它根据解码的上下文从数据存储中检索 k 个最近邻候选标记，并将其处理为概率。最后，它通过对模型概率和检索概率进行加权融合来校准翻译模型的概率分布。 k NN-MT 在跨领域和多语言机器翻译任务中，与传统模型相比，展现出了更优越的迁移性和泛化能力。后续研究集中于提升其性能和效率 [301, 367, 421]，或将其方法应用于其他自然语言处理任务 [20, 312]。

测试时计算综述：从直觉推理到深思熟虑的推理 7

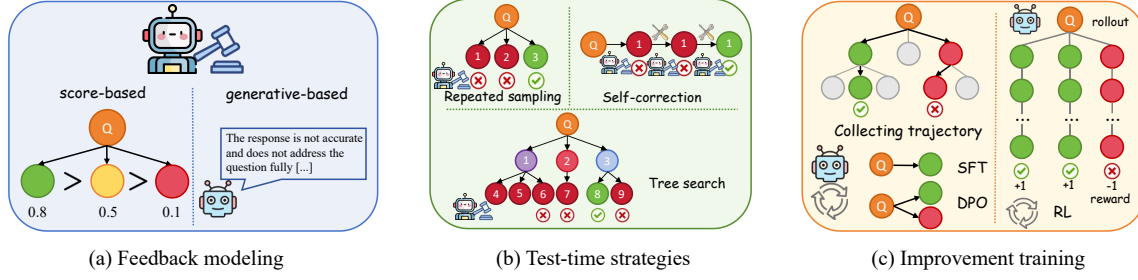


图 4. 测试时推理中反馈建模、搜索策略和改进训练的示意图。

Summary 1: *Parameter updating and output calibration are the most versatile TTA methods. However, parameter updating suffers from training instability and inefficiency in LLMs, while output calibration relies on target domain information and risks knowledge leakage. Input modification and representation editing are free from training but have limited applicability: input modification is related to ICL capabilities, and representation editing demands manual prior knowledge.*

4 面向系统 2 思维的测试时推理

Test-time reasoning aims to spend more inference time to search for the most human-like reasoning process within the 庞大的解码搜索空间。本节将介绍测试时推理的两个核心组成部分：反馈建模与搜索策略（如图 4 所示）。

4.1 反馈建模

基于分数的反馈。基于分数的反馈，也称为验证器，旨在预测标量分数以评估生成结果与真实答案或人类认知过程的对齐程度。其训练过程通常类似于基于人类反馈的强化学习中的奖励模型，利用各种形式的反馈信号，并将其建模为分类 [50] 或排序任务 [22, 108, 375]。在推理任务中，验证器主要分为两类：基于结果的验证器和基于过程的验证器。基于结果的验证器 [50] 使用最终思维链结果的正确性作为训练信号。

刘等人 [196] 提供了训练强基于结果的验证器的详细方法。相比之下，基于过程的验证器 [179, 296, 390] 基于每个推理步骤的正确性进行训练。与基于结果的验证器相比，基于过程的验证器不仅能评估中间推理步骤，还能更精确地评估整个推理过程。然而，基于过程的验证器需要更多的人工努力来标注中间步骤的反馈。数学牧羊人 [311]、欧米茄基于过程的验证器 [206] 和史诗基于过程的验证器 [282] 利用蒙特卡洛树搜索算法自动收集高质量的过程监督数据。张等人 [407] 利用评论家模型评估蒙特卡洛树搜索收集的过程标注，过滤低质量数据以提高基于过程的验证器的训练有效性。塞特卢尔等人

[258] 认为基于过程的验证器应评估每个步骤对后续推理的优势，而非仅关注其正确性。他们提出过程优势验证器，并通过蒙特卡洛树搜索高效构建训练数据。此外，卢等人 [201] 和袁等人 [376] 注意到基于结果的验证器隐式地建模了每个步骤的优势，因此他们分别利用基于结果的验证器自动标注过程监督数据，或直接在结果标签上训练基于过程的验证器。刘等人 [193] 根据具有更高不确定性的标记自适应地分割推理步骤，用于训练更好的基于过程的验证器。

Category	Sub-category	Representative Methods	Domain	Objective	Description
Score-based	ORM	Cobbe et al. [50]	Math	classification	ORM; human annotated data
		Acemath [196]	Math	list-wise Bradley-Terry	ORM; sampling training data from multiple LLMs
	PRM	Lightman et al. [179]	Math	classification	PRM; human annotated data
		Math-shepherd [311]	Math	classification	PRM; annotating processes via MC estimation
		Zhang et al. [407]	Math	classification/regression	PRM; annotating processes via MCTS and LLM-as-a-judge
		Implicit PRM [376]	Math	implicit reward modeling	PRM; training PRMs with outcome labels
Generative-based	Training-free	ASPRM [193]	Math, Code	classification	PRM; adaptive segment step; annotating processes via MC estimation
		LLM-as-a-Judge [414]	General	-	Designing system instructions to mitigate biases
	Training-based	BSM [250]	General	-	Dividing into multiple criteria and then merging
		Shepherd [314]	General	SFT	Collecting data from human annotation and the Internet
		Prometheus [140]	General	SFT	Training single and pairwise models and then merging them
		EvalPlanner [251]	General	DPO	Planing evaluation processes and then evaluating
		GenRM [394]	Math	SFT	PRM; synthesizing critique data from external LLMs
		R-PRM [260]	Math	SFT & DPO	PRM; synthesizing critique data from external LLMs
		Critic-RM [372]	General	SFT & Bradley-Terry	ORM; synthesizing and filtering critique data via self-critique
		CLoud [8]	General	SFT & Bradley-Terry	ORM; synthesizing data from external LLMs and self-critique

表 1. 反馈建模方法概览。

基于生成的反馈。尽管验证器能够评估生成答案或步骤的正确性，但其缺乏可解释性，无法定位错误的具体原因或提供修正建议。

基于生成的反馈，也称为批评者，充分利用了大语言模型的生成与指令遵循能力 [8, 48, 260, 364]。通过设计特定指令，它能够从多个维度进行逐点或成对评估，甚至以自然语言提供修改建议。强大的闭源大语言模型，如 GPT-4 和 Claude，是有效的批评者。它们能够对生成文本进行详细且可控的评估，例如事实性、逻辑错误、连贯性和对齐性，与人工评估具有高度一致性 [47, 191, 208, 308]。

然而，它们仍面临长度、位置和困惑度等偏差 [19, 275, 310]。大模型作为评审 [414] 精心设计系统指令以减轻偏差干扰。BSM[250] 基于多个标准进行评估，然后进行合并。Peng 等人 [233] 采用多智能体联合评估答案的事实性和指令遵循性。

为了获得更廉价的基于语言的反馈，开源大语言模型（LLM）也可以通过监督微调（SFT）成为有竞争力的替代方案 [178, 231, 323, 420]。Shepherd[314] 从人工标注和在线社区收集高质量训练数据，以微调评估模型。Auto-J[159] 从各种场景收集查询和响应，并为每个场景设计评估标准。GPT-4 随后根据这些标准生成对响应的批评，并将其批评能力蒸馏到开源大语言模型中。Prometheus[139, 140] 设计了更细粒度的评估维度。它分别训练一个评估模型和一个成对排序模型，然后通过权重合并将它们统一到一个大语言模型中。为了减少对人工标注和外部大语言模型的依赖，Wang 等人 [313] 提出了一种自训练方法：批评模型生成正负响应，然后通过拒绝采样收集批评数据以进行迭代微调。在自训练的基础上，EvalPlanner[251] 使批评模型能够规划评估过程 and 标准，基于这些进行批评，然后收集正负样本以通过 DPO 改进批评模型 [245]。[370] 精心合成和过滤数据，使基础模型仅用 40K 样本进行 SFT 和 DPO 训练就能获得强大的批评能力。GenRM[394] 利用指令微调使验证器能够回答“答案是否正确（是/否）？”，并使用生成“是”标记的概率作为分数。GenRM 还可以结合思维链（CoT），允许验证器在回答“是”或“否”之前生成相应的推理过程。ThinkPRM[131] 使用长思维链评估每个推理步骤，与判别式 PRM 相比，仅需 1% 的过程监督数据。Critic-RM[372] 联合训练批评器和验证器。

Category	sub-category	Representative Methods	Tasks	Verifier/Critic	Train-free
Repeat Sampling	Majority voting	CoT-SC [318]	Math, QA	self-consistency	✓
		PROVE [290]	Math	compiler	✓
	Best-of-N	Cobbe et al. [50]	Math	ORM	✗
		DiVeRSe [172]	Math	PRM	✗
Self-correction	Human feedback	Knockout [195]	Math	critic	✓
		NL-EDIT [64]	Semantic parsing	Human	✗
	External tools	FBNET [285]	Code	Human	✗
		DrRepair [360]	Code	compiler	✗
		Self-debug [40]	Code	compiler	✓
	External models	CRITIC [88]	Math, QA, Detoxifying	text-to-text APIs	✓
		REFINER [231]	Math, Reason	critic model	✗
		Shepherd [314]	QA	critic model	✗
		Multiagent Debate [63]	Math, Reason	multi-agent debate	✓
		MAD [177]	Translation, Math	multi-agent debate	✓
Tree Search	Uninformed search	Self-Refine [214]	Math, Code, Controlled generation	self-critique	✓
		Reflexion [263]	QA	self-critique	✓
	Heuristic search	RCI [135]	Code, QA	self-critique	✓
		ToT [359]	Planing, Creative writing	self-critique	✓
		Xie et al. [342]	Math	self-critique	✓
		RAP [99]	Planing, Math, Logical	self-critique	✓
		TS-LLM [69]	Planing, Math, Logical	ORM	✗
		rStar [238]	Math, QA	multi-agent consistency	✓
		ReST-MCTS* [387]	Math, QA	PRM	✗

表 2. 搜索策略概览。

4.2 搜索策略

4.2.1 重复采样。

诸如 top-p 和 top-k 的采样策略是大语言模型推理中常用的解码算法。它们在解码过程中引入随机性以增强文本多样性，从而允许并行采样多个生成文本。

通过重复采样，我们获得更多机会找到正确答案。重复采样特别适用于可自动验证的任务，例如代码生成，其中我们可以利用单元测试轻松地从一个或多个样本中识别出正确解 [169, 248]。对于难以验证的任务，如数学应用题，重复采样有效性的关键在于验证策略。

验证策略。验证策略包括两种类型：多数投票和最佳 N 选（Best-of-N, BoN）采样。

多数投票 [160, 180] 选择样本中出现频率最高的答案作为最终答案，其动机源于集成学习。多数投票简单而有效。例如，自洽性思维链（Self-Consistency CoT）[318] 在数学推理任务中比普通思维链（CoT）的准确率提高了 18%。然而，多数并不总是掌握真理，因为它们可能犯类似的错误。因此，一些研究在投票前进行过滤或加权。例如，PROVE 框架 [290] 将思维链转换为可执行程序，若程序结果与推理链结果不一致则过滤掉该样本。Kang 等人 [130] 提出自确定性度量，根据排名对投票进行加权。Huang 等人 [116] 使用校准置信度作为投票权重，并根据响应一致性调整样本数量以提高效率 [1]。RPC [419] 从理论和实验上证明，过滤低概率样本可提高多数投票的性能和效率。

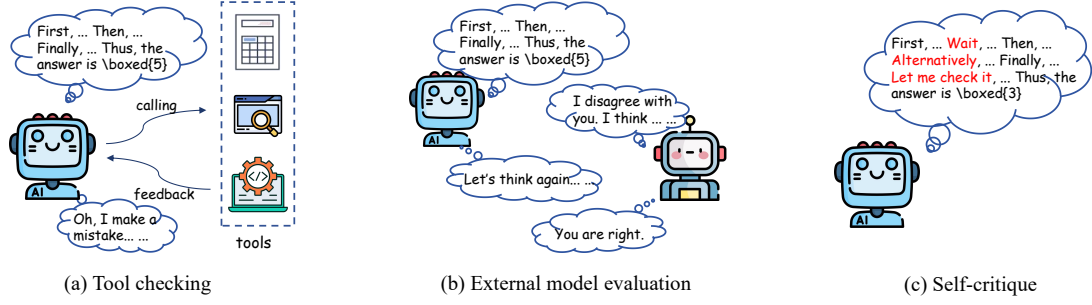


图 5. 带有工具检查、外部模型评估和自我批判的自我纠正示意图。

Best-of-N sampling uses a verifier to score each response and selects the one with the highest score as the final answer [50, 184, 221, 273]. Li et al. [172] propose a voting-based BoN variant, which performs weighted voting on all answers based on the verifier's scores and selects the answer with the highest score. [195] design BoN in a knockout tournament, using pairwise comparison verifiers to filter out the best response. In addition, some works aim to improve the efficiency of BoN. Inspired by speculative decoding, Qiu et al. [241], Sun et al. [281], Yu et al. [368], Zhang et al. [397] and Manvi et al. [215] evaluate each reasoning step and prune low-scoring sampled results, halting further generation for those paths, thereby significantly reducing the overall time cost. PRS [362] enables LLMs to self-critique and self-correct, guiding the model to generate expected responses with fewer sampling times. Li et al. [156] compare BoN and majority voting, demonstrating that BoN is suitable for harder questions with moderate diversity of response distribution.

改进训练。重复采样已被证明是一种简单而有效的方法，甚至超越了使用 RLHF 微调的模型 [80, 109]。然而，它消耗大量推理时间，在实际应用中难以承受。

因此，许多研究尝试通过最优 N 采样 (BoN Sampling) 训练模型以逼近最优 N 分布 (BoN Distribution)，从而减少推理时的搜索空间。ReST [94] 从策略模型中采样奖励值高于阈值的响应作为自训练数据，并通过离线强化学习微调策略模型。在每次迭代中，ReST 采样新的训练数据。vBoN [6]、BoNBoN [93] 和 BOND [257] 推导最优 N 分布，并最小化策略模型分布与最优 N 分布之间的差异。Chow 等人 [49] 设计了一种最优 N 感知损失，使策略模型在微调过程中更具探索性。

4.2.2 自我纠正。

自我纠正是一种顺序测试时计算方法，使大语言模型能够基于外部或内部反馈迭代地修订和优化生成结果 [263]。

反馈来源。用于自我纠正的反馈通常以自然语言呈现，来源多样，包括人工评估、工具检查、外部模型评估和内在反馈。人工评估是反馈的黄金标准，但由于其高成本和有限的可扩展性，主要用于早期研究以探索自我纠正能力的上限 [64, 284, 285]。对于某些特定领域的任务，外部工具检查提供准确且高效的反馈 [40, 79, 88]。例如，Yasunaga 和 Liang [360] 提出在代码修复和生成任务中从编译器获取反馈。在具身任务中，环境可以为基于大语言模型的智能体的动作轨迹提供精确反馈 [304]。

外部模型评估是通用任务的有效反馈来源，例如第 4.1 节中描述的各种基于语言的批评模型。例如，Paul 等人 [231] 首先为自然语言推理任务定义了多种错误类型，然后设计了相应的反馈模板。他们使用合成的反馈训练数据训练了一个评估模型，借助该批评模型，推理模型实现了显著的性能提升。多智能体辩论 [34, 63, 177, 309, 344] 是另一种利用外部反馈增强推理能力的机制。在该方法中，模型没有明确的推理者和批评者角色区分。相反，多个模型独立进行推理，相互批评，并根据反馈捍卫或改进自己的推理。这一过程持续进行，直到智能体达成共识或由裁判模型总结最终推理结果。多智能体辩论在事实核查 [132, 137]、常识问答 [344]、忠实性评估 [28] 和复杂推理 [63] 方面展现了潜力。

然而，多智能体辩论可能不稳定，因为大语言模型容易受到对抗性信息的影响，并可能在误导性输入下将正确答案修改为错误答案 [5, 144]。因此，成功的多智能体辩论要求大语言模型在面对其他模型的错误答案时保持立场，同时对有效建议保持开放态度 [272]。一般来说，参与辩论的大语言模型越多，整体推理性能越强。

然而，这显著增加了所需的大语言模型推理次数和输入上下文长度，对大语言模型推理成本构成了重大挑战 [190]。为了降低辩论推理成本，Li 等人 [170] 研究了多个智能体之间拓扑连接的影响，并表明稀疏连接（如环形结构）并不逊色于全连接拓扑。GroupDebate[190] 将大语言模型分组，在组内进行辩论，仅在组间共享共识结果。

自我批评假设大语言模型能够自我评估其输出，并通过内在反馈优化输出 [379]。

这一思想源于计算复杂度理论中的一个基本原理：验证一个解是否正确通常比求解问题本身更容易。Bai 等人 [16] 提出通过提示自身来纠正大语言模型（LLM）生成的有害响应。自精炼（Self-Refine）[214] 和 RCI 提示（RCI Prompting）[135] 在算术推理等任务中迭代地提示大语言模型自我纠正其响应。IoE[162] 观察到，大语言模型在自我批评过程中可能过度批评自身，导致性能下降，并设计了提示来引导大语言模型评估置信度。

ProgCo[268] 利用代码在表达复杂逻辑方面的优势，使大语言模型能够以伪代码形式生成响应，随后进行自我批评与精炼。SETS[33] 结合了重复采样与自我批评的优势，对每条采样得到的推理路径应用自我批评与纠正，并通过多数投票选择最终解。S1[218] 在推理过程中加入“等待”，促使大语言模型批评其推理过程。

Arguments. The effectiveness of self-correction, especially the self-critique, has remained controversial. Several empirical studies on code generation [225], commonsense QA [117], math problem-solving [307], planning [297], and graph coloring [270] confirm that self-correction is not a guaranteed solution for improving performance. Kamoi et al. [128] 认为自我纠正的有效性被高估了。以往的成功要么依赖于预言答案，要么依赖于较弱的初始答案。只有那些可以分解为易于验证的子任务的任务才能真正从自我纠正中受益。他们建议微调特定的评估模型以实现更好的自我纠正。Zhang 等人 [396] 尝试通过类人的认知偏差来解释和缓解自我批评的失败。Tyen 等人 [295] 解耦了大语言模型识别错误和纠正错误的能力，并创建了相应的评估数据集。评估结果表明，大语言模型在自我纠正过程中并不缺乏纠正错误的的能力，其主要性能瓶颈在于定位错误。Yang 等人 [357] 将自我批评分解为置信度和批评能力。实证研究表明，微调对于同时增强这两种能力是必要的，而提示工程只能实现一种权衡。

改进训练 上述大多数自我纠正方法在先进的闭源大模型或参数超过 700 亿的开源大语言模型上表现出显著的性能提升。然而，对于能力较弱的中等规模开源模型，我们需要进一步微调以解锁其自我纠正能力。监督微调利用高质量的多轮纠正数据优化模型，这些数据可以来自人工标注 [252]、自我合理化 [378, 381]、多智能体辩论 [280] 或从更强的大语言模型中采样 [7, 78, 231, 243, 335, 404]。GLoRe[103] 认为大语言模型需要针对不同类型的错误进行全局或局部细化。为此，他们构建了全局和局部细化的训练集，训练验证器识别全局和局部错误，并开发基于不同全局或局部反馈信号进行细化的大语言模型。Xi 等人 [335] 设计了一个可扩展的框架来合成自我纠正训练数据，使推理模型能够生成受控错误并接收批评者的反馈进行自我纠正。尽管监督微调有效，但离线生成的自我纠正轨迹训练数据只能模拟有限的纠正模式。这导致与模型推理过程中实际自我纠正行为的分布不匹配。Self-correct[327] 采用在线模仿学习，在每个训练轮次后重新采样新的自我纠正轨迹用于训练。为了进一步扩大大语言模型的探索空间，许多研究采用灵活的强化学习算法以超越监督微调的性能极限。SCoRe[143] 提出使用多轮强化学习方法提升自我批评和自我纠正能力。T1[110] 采用自我纠正训练数据进行监督微调冷启动，随后使用 RLOO 算法 [3] 进行强化学习训练。在强化学习阶段，高温采样和熵奖励鼓励大语言模型探索更多样化的推理路径。Deepseek-R1[95] 使用基于规则的奖励和 GRPO 算法 [259] 进行强化学习训练。它还展示了强化学习的巨大潜力，即使没有监督微调冷启动，其探索能力也足以赋予大语言模型强大的推理能力。

4.2.3 树搜索。

重复采样和自我纠正分别以并行和顺序的方式扩展测试时计算。人类思维是一种树搜索，它将并行头脑风暴与在遇到死胡同时回溯寻找其他解决路径相结合。搜索算法和价值函数是树搜索中的两个关键组成部分。

搜索算法。在大语言模型推理中，当前的搜索算法包括无信息搜索和启发式搜索。

无信息搜索根据固定规则探索搜索空间。例如，思维树（ToT）[359] 采用广度优先搜索或深度优先搜索进行搜索，而 Xie 等人 [342] 使用束搜索。对于搜索空间较大的问题，无信息搜索通常效率较低，因此以 A^* [216, 300] 和蒙特卡洛树搜索（MCTS）[21, 99, 230] 为代表的启发式搜索策略在推理任务中被广泛使用。蒙特卡洛树搜索无需显式启发式，利用随机仿真和不确定环境下的自适应树扩展，非常适合大规模状态空间。它通过选择、扩展、仿真和反向传播四个步骤逐步优化搜索结果，逼近最优解。相比之下， A^* 使用启发式函数引导的确定性搜索来保证最优路径，但其性能取决于启发式函数的设计。因此，蒙特卡洛树搜索已成功应用于检索增强生成（RAG）[68, 114, 124, 158]、问答（QA）[73, 203]、幻觉缓解 [46]、文本到 SQL[377] 等任务。此外，Long[198] 使用强化学习训练大语言模型控制器来引导大语言模型推理器的搜索路径，Chari 等人 [29] 利用蚁群进化算法引导树搜索。

价值函数。价值函数评估每个状态的价值，并在启发式树搜索中引导树向具有更高价值的分支扩展。Xu[348] 通过噪声对比估计训练能量函数作为价值函数。RAP[99] 设计了一系列启发式价值函数，包括动作的似然、状态的置信度、自我评估结果和任务特定奖励，并根据任务需求将它们组合起来。可靠

广义价值函数促进了蒙特卡洛树搜索在搜索空间更深、更复杂问题上的应用。AlphaMath [31] 与 TS-LLM [69] 用学习到的 LLM 价值函数替代手工设计的价值函数，在蒙特卡洛树搜索中自动生成推理过程和步骤级评估信号。VerifierQ [236] 融合隐式 Q 学习与对比 Q 学习来训练价值函数，有效缓解了步骤级的过估计问题。传统蒙特卡洛树搜索方法仅扩展一条轨迹，而 rStar [238] 认为当前价值函数难以准确引导最优路径的选择。因此，rStar 保留多条候选路径，并用另一个 LLM 进行推理，最终选择两个 LLM 推理结果一致的路径。Gao 等人 [81] 提出 SC-MCTS，结合对比奖励、似然和自我评估等多个奖励模型作为价值函数。MCTS_r [385] 与 SR-MCTS [386] 将完整响应作为节点，通过自我批评与修正扩展搜索空间。SR-MCTS 利用成对偏好奖励和全局分位数得分作为价值函数，提供了更稳健的价值函数估计。

改进训练 树搜索可以引导 LLM 生成推理过程，这些数据有助于训练推理能力更强的 LLM [92, 346, 383]。ReST-MCTS* [387] 使用过程奖励作为价值函数来引导蒙特卡洛树搜索，收集高质量推理轨迹和每一步的价值，以改进策略模型和奖励模型。由于树搜索的逐步探索，可以获得更细粒度的步骤级反馈信号。

MCTS-DPO [341] 通过蒙特卡洛树搜索收集步骤级偏好数据，并使用 DPO 进行偏好学习。AlphaLLM-CPL [316] 基于偏好奖励差距和策略预测差距对轨迹进行排序，采用课程学习高效利用蒙特卡洛树搜索收集的轨迹。近期，许多 LMR [240, 403, 412] 也证实了使用树搜索构建高质量长推理链数据进行训练的必要性。

总结 2：重复采样易于实现，并能提升答案多样性，适用于开放式或易于验证的任务，但计算效率较低。自我纠正依赖于精确、细粒度的反馈，在易于验证的任务上表现良好，但在反馈质量差或推理能力较弱时可能效果不佳。树搜索能够全局优化复杂的规划任务，但实现较为复杂。

5 高级主题

5.1 可泛化的系统 2 模型

目前，大多数大推理模型在数学和代码等特定领域表现出强大的推理性能，但在跨领域、跨语言或通用任务上的泛化能力仍然不足 [266, 411]。一方面，作为当前系统 2 模型的基础，思维链在非符号推理任务中效果有限 [269]。另一方面，验证器和批评器的泛化能力有限，难以为推理模型提供有效指导 [35, 138, 154]。针对前一个局限性，集成外部工具和多智能体反馈是一个有前景的方向 [147, 222]。许多工作旨在使推理模型在推理过程中学会执行检索或工具使用。Search-o1 [164] 通过生成表示搜索动作的特殊标记来触发外部搜索工具，而 Search-R1 [126] 则通过强化学习进一步增强大型语言模型与搜索工具之间的交互。

深度研究 [87, 166, 228, 415] 集成了网络搜索和代码执行等多种工具，在信息综合、报告撰写和复杂代码生成等通用任务上表现出卓越的性能。

针对后一个挑战，一些工作利用多目标训练 [305]、模型集成 [181]、软奖励 [279] 或正则化约束 [123, 354] 来提升验证器的泛化能力。与判别式验证器相比，生成式批评器继承了大型语言模型的泛化能力，并能够利用测试时计算实现更大的

14 Ji 等人。

潜力 [141]。GenPRM[409] 和 GRM[197] 利用重复采样进行测试时计算，GRM 训练一个元奖励模型来评估批评器输出的质量。RM-R1[39] 和 JudgeLRM[37] 使用强化学习训练，使批评器具备自我纠正能力。

此外，弱到强泛化（weak-to-strong generalization）[25] 是一个值得进一步探索的课题。人们不再满足于解决具有标准答案的数学问题；他们希望系统 2 模型能够辅助科学发现和数学猜想的证明。在这种情况下，即使是人类专家也难以提供准确的反馈，而弱到强泛化为解决这一问题提供了一个有前景的方向 [287]。

5.2 多模态测试时计算

视觉、音频、视频等多种模态对于模型理解和与世界交互至关重要。与大型语言模型（LLM）类似，大型多模态模型（LMM）在扩展训练计算方面也面临挑战。因此，为多模态任务实现高效的测试时扩展已成为一个前沿研究课题。在系统 1 思维中，测试时增强（TTA）已成功应用于 LMM，提升了零样本图像分类、图文检索和图像描述生成等任务的性能 [410]。对于多模态推理任务，多模态思维链（CoT）[77, 150, 217, 332, 398, 405] 以及多模态批评器或验证器 [293, 345] 的探索为构建多模态系统 2 模型开辟了可能性。例如，IoT [418] 和 SketchPAD [115] 利用视觉工具获取关键视觉信息，并将其整合到 CoT 中以优化多模态推理。VisualPRM [315] 通过蒙特卡洛估计构建过程标注数据，并训练轻量级过程奖励模型（PRM）验证器；而 GPT-4o 和 Qwen-VL 等前沿 LMM 可以作为批评器。在这些基础上，重复采样 [182] 和树搜索 [60, 358] 能够有效进一步提升 LMM 的多模态推理性能。Xu 等人 [347] 将视觉推理过程划分为四个阶段：任务总结、图注、推理和答案结论。他们提出了一种阶段级束搜索方法，在每个阶段重复采样并选择最佳结果进入下一阶段。Audio-Reasoner [343] 将四阶段 CoT 框架迁移到音频模态。为了使 LMM 具备自我纠正能力，Deepseek-R1 的训练方案也已在 LMM 中得到实践 [32, 213, 261]。OpenVLThinker [58] 和 Vision-R1 [118] 将图像转换为文本图注，使纯文本大型推理模型（LRM）能够融入视觉信息以生成 CoT。这作为进一步强化学习训练多模态推理模型的冷启动数据。LMM-R1 [234] 表明，通过纯文本强化学习获得的推理能力可以作为多模态强化学习训练的有效冷启动。

除了理解和推理任务，多模态生成任务也能从测试时计算中受益。

主流多模态生成模型分为基于扩散的 [232, 356] 和基于自回归的 [289, 340] 模型。虽然扩散模型缺乏显式的思维链，但它们可以在测试时通过采样各种高斯噪声 [339] 来扩展搜索空间。对于视频生成，由于重复采样完整视频的计算成本很高，Cong 等人 [51] 使用过程奖励模型逐帧验证，并应用束搜索来剪枝低分帧。Liu 等人

[185] 提出帧树搜索，使用评论家在不同标准下评估视频的早期、中期和后期阶段，每一步只保留得分最高的前 k 帧。自回归模型将中间生成的图像视为推理步骤。Guo 等人 [96] 提出潜在评估奖励模型（PARM）来评估这些中间图像的质量潜力，并引入反思机制，使生成模型能够自我纠正低质量图像。

5.3 高效的测试时计算

测试时计算的成功应用表明，牺牲推理效率可以带来更好的推理性能。然而，研究人员继续寻求性能与效率之间的平衡，旨在实现

在固定推理延迟预算下的最优性能。这需要为每个样本自适应地分配计算资源。Damani 等人 [55] 训练一个轻量级模块来预测问题的难度，并根据其难度分配计算资源。Zhang 等人 [393] 进一步将分配目标扩展到更多超参数。Chen 等人 [41] 和 Wang 等人 [320] 系统地评估了大型推理模型中的过度思考和思考不足现象，前者导致模型对简单问题过度复杂化，后者导致在困难问题上频繁切换推理路径，从而降低推理效率。这些现象甚至导致大型推理模型在处理简单问题时表现不佳 [399]。仍有许多开放问题值得探索，例如如何将推理加速策略（如模型压缩 [120, 165, 176]、令牌剪枝 [71, 389] 和推测解码 [153, 337]）与测试时计算相结合，以及如何根据问题难度分配最优推理预算 [45, 98, 306]。

长链思维（Long CoT）会导致推理延迟和键值缓存（key-value cache）的内存占用。近期，大量研究探索了多种策略以缩短推理长度。早期工作主要聚焦于使用变长思维链数据的监督微调（SFT）模型 [52, 129, 189, 211, 220, 336, 355, 371]。相比之下，主流方法将基于长度的奖励纳入强化学习（RL）[11, 205, 288, 365, 366, 374]，以鼓励简洁的响应。肖等人 [338] 度量了视觉-语言推理任务的复杂度，并设计了一种复杂度感知的课程学习与 GRPO 算法。他们按照从难到易的顺序组织训练样本，并根据任务复杂度动态调整奖励函数和 KL 项权重。L1[2] 与弹性推理 [350] 能够基于给定的令牌预算实现对响应长度的精确控制。此外，自预算器 [173] 能够在生成前自主估计最优令牌预算，并允许用户选择是等待完整响应还是在生成过程中提前终止。然而，监督微调和强化学习都需要大量的计算资源。若干研究转而探索免训练的高效推理方法，包括提示 [14, 98, 149, 349]、模型合并 [288, 330] 和提前退出 [72, 210, 353]。值得注意的是，DEER[353] 在思维链生成过程中基于中间答案的置信度做出提前退出决策，有效降低了推理开销。随后，戴等人 [54] 引入了一种新的强化学习范式，鼓励模型在适当的时候激励提前终止思考。另一种实现效率的有前景的方法是混合推理，它根据任务复杂度在简洁响应和长链推理之间动态切换。这些方法 [65, 125, 199, 204, 391] 遵循类似的流程：它们首先在长短思维链样本混合数据集上进行监督微调作为冷启动，然后改进策略优化算法以训练模型获得混合推理能力。

6 未来方向

6.1 测试时缩放定律

与训练时计算规模的扩展不同，测试时计算仍缺乏通用的缩放定律。一些工作尝试为特定的测试时计算策略推导缩放定律 [152, 333]。Brown 等人 [23] 证明，性能与重复采样次数近似呈对数线性关系。Chen 等人 [43] 将重复采样建模为淘汰赛和联赛式算法，从理论上证明重复采样的失败概率遵循幂律缩放。Snell 等人 [265] 研究了重复采样和自我纠正的缩放定律，并提出了计算最优的缩放策略。实现通用缩放定律面临两大挑战：首先，当前的测试时计算策略多种多样，每种策略通过不同机制引导模型，因此缺乏一个通用框架来描述它们；其次，测试时计算的性能

受到多种因素的影响，包括样本难度、反馈信号的准确率以及解码超参数，我们需要实证研究来筛选出关键因素。

6.2 策略组合

不同的测试时计算策略适用于不同的任务和场景，因此组合多种策略是实现更好的系统 2 思维的一种方式。例如，Marco-o1[412] 结合了蒙特卡洛树搜索（MCTS）和自我纠正，使用 MCTS 规划推理过程，并利用自我纠正提高每一步的准确率。TPO[171] 结合了最优 N 采样（BoN 采样）和自我纠正。此外，系统 1 模型中的测试时自适应策略也可以与测试时推理策略相结合。Akyürek 等人 [4] 将测试时训练与重复采样相结合。他们进一步优化测试样本上的语言建模损失，然后通过数据增强生成多个候选答案，最后通过多数投票确定答案。他们展示了测试时训练在推理任务中的潜力，在 ARC 挑战中超越了人类平均水平。因此，我们认为对于大语言模型（LLM）推理，不仅需要关注新兴的测试时策略，还需要关注测试时自适应方法。通过有效组合这些策略，我们可以开发出达到或超越 o1 级别性能的系统 2 模型。

6.3 新的测试时计算范式

潜空间思维链推理已成为测试时计算的一种有前景的范式。通过在潜空间中进行内部推理，它克服了传统显式思维链在计算效率和抽象推理能力方面的局限性 [89, 262, 351, 352]。椰子模型 [100] 通过直接将模型的最终隐藏状态作为下一步输入嵌入，在连续潜空间中进行推理，从而产生了高级推理模式。压缩思维链 [45] 引入了一个框架，生成连续且可变量度的沉思标记作为推理链的压缩表示，实现了高效且准确的推理。轻思考者 [392] 通过专门的数据构建和注意力掩码设计，将中间推理步骤压缩为紧凑的要点标记，从而提高了大语言模型的效率，使模型能够丢弃冗长的推理链，同时保持对压缩表示进行推理的能力。潜空间思维链目前面临若干挑战，包括直接监督的困难、泛化能力有限以及对可解释性的担忧 [42]。这要求进一步探索标记级策略和内部机制，以实现更鲁棒和透明的潜空间推理。

7 基准

测试时自适应。在系统 1 模型中，分布偏移包括对抗鲁棒性、跨领域和跨语言场景。在计算机视觉领域，ImageNet-C[107]、ImageNet-R[105]、ImageNet-Sketch[303] 是测试时自适应的常用数据集。Yu 等人 [373] 提出了一个基准，用于在 5 个图像分类数据集上对不同测试时自适应设置和骨干网络下的测试时自适应方法进行统一评估。对于自然语言处理任务，测试时自适应主要应用于问答和机器翻译任务，常用数据集包括 MLQA[155]、XQuAD[12]、MRQA[70] 和 CCMatrix[256]。

反馈建模。奖励基准 [146] 收集了 20.2k 个提示-选择-拒绝三元组，涵盖对话、推理和安全等任务。它评估奖励模型区分所选响应和拒绝响应的准确率。

奖励模型基准 [194] 进一步评估了响应风格对奖励模型的影响。奖励模型基准 [417] 将评估扩展到更实用的最优 N 设置，其中奖励模型需要从多个候选中选择最佳响应。

批评家基准 [183] 专门设计用于评估批评模型的生成、批评和纠正能力。

For PRM, Song et al. [267] propose PRMBench, which evaluates PRMs whether they can identify the earliest incorrect

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数学任务中的推理步骤。过程基准（ProcessBench）[413] 提供了更细粒度的评估，包括冗余性、可靠性和敏感性。此外，还有用于评估多模态反馈建模的基准，如视觉语言奖励基准（VL-RewardBench）[163]、视觉常识推理基准（VCR-Bench）[237] 和多模态判断基准（MJ-Bench）[44]。

测试时推理。推理能力是系统二模型的核心，涵盖数学、代码、常识、规划等 [382]。数学推理是最引人注目的推理任务之一。随着大语言模型和测试时计算的进步，一些此前具有挑战性的基准（如 GSM8K [50] 和 MATH [106]）上的准确率已超过 90%。因此，更难的大学生入学考试 [10, 15, 400] 和竞赛级 [75] 数学基准被提出。一些竞赛级基准不局限于代数、逻辑推理和应用题中的文本模式。例如，奥林匹克基准（OlympiadBench）[104]、奥林匹克竞技场（OlympicArena）[119] 和美国数学邀请赛（AIME）[380] 为几何问题提供图像，融入视觉信息以辅助解题，而阿尔法几何（AlphaGeometry）[291] 采用符号规则进行几何证明。目前最具挑战性的基准是前沿数学（FrontierMath）[84]，其问题由数学家精心设计，涵盖现代数学的主要分支。即使是最先进的 o3 模型也未达到 30% 的准确率。

代码能力是大语言模型推理的关键方面，具有很高的实用价值，涵盖代码补全 [59, 85, 388]、代码推理 [90, 325] 和代码生成 [13, 36] 任务。其中，代码生成受到更多关注。

人类评估（HumanEval）[36] 和多语言编程基准（MBPP）[13] 提供编程问题的自然语言描述，要求大语言模型生成相应的 Python 代码并使用单元测试进行评估。多语言扩展（MultiPL-E）[27] 将它们扩展到 18 种编程语言。评估增强（EvalPlus）[187] 自动扩充测试用例以评估生成代码的鲁棒性。最近，一些研究从开源项目中收集基准，这些基准更接近实际应用，并且由于复杂的函数调用而更具挑战性，例如数据科学基准（DS-1000）[145]、代码评估（CoderEval）[369]、进化代码基准（EvoCodeBench）[157] 和大代码基准（BigCodeBench）[424]。

常识推理要求大语言模型同时具备常识与推理能力。策略问答（StrategyQA）[83] 收集了复杂且微妙的多跳推理问题。大规模多任务语言理解（MMLU）[106] 和大规模多任务语言理解专业版（MMLU-Pro）[322] 涵盖了多个领域的常识推理问题，包括科学、技术、工程和数学（STEM）、人文学科、社会科学等。规划旨在使大语言模型能够根据当前状态和环境采取最优行动以完成任务。当前的规划基准主要集中于合成任务，例如积木世界（Blocksworld）[298]、填字游戏和 24 点游戏（Game-of-24）[359]。

8 结论

本文对现有测试时计算的相关工作进行了全面综述。我们介绍了系统一（System-1）和系统二（System-2）模型中的各种测试时计算方法，并展望了该领域未来的发展方向。

我们认为测试时计算能够帮助模型更好地处理复杂的现实世界分布与任务，是推动大语言模型迈向认知智能的一条有前景的路径。希望本文能够促进进一步的研究。

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